Bayesianism for the Indecisive[†]

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Measure of Progress

One measure of progress in Bayesian epistemology is the range of non-Bayesians who can be convinced by Bayesian arguments. I look at a non-Bayesian who is not persuaded by existing arguments and develop an argument to convince them.

Indecisive Agents

Call an agent indecisive if their opinions about how events compare do not form a complete order.

The non-Bayesian I want to convince is an indecisive agent with a policy or dispositions to act that cannot be represented as Bayesian.

Example: Geordi is almost late. His train is either on the North or the South platform. He can go to the North platform, the South platform, or the bus stop. He has no opinion about which platform is more likely the right one or even if they're equally likely. The losses are:

	North	South	Bus
North Correct	0 min late	30 min late	20 min late
South Correct	30 min late	0 min late	20 min late

He decides to take the bus. This is not a Bayes solution.

Existing Arguments

Dutch Book Arguments	X
Accuracy Arguments	X
Waldian Argument (see below)	X

The Waldian Argument: Since Geordi's policy is not a Bayes solution, by Wald's Complete Class Theorem we know there is a Bayes solution that risk dominates his policy. So, he should change his mind.

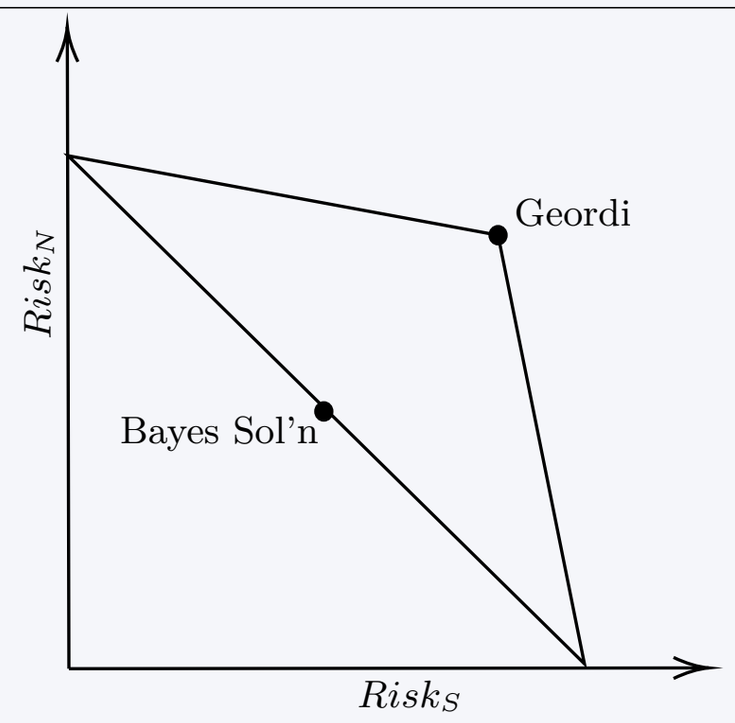
Rebuttal to the Waldian Argument

Geordi concedes that there is a policy that risk dominates his policy, but points out that this argument doesn't show which one it is. Wald's Complete Class Theorem is incomputable, so there may be no way for us to find it. Switching to an arbitrary Bayes solution might increase his risk, so he will stick to his policy.

Computable Waldian Argument

When the context is restricted (see below), there is a computable procedure that will find the dominating Bayes solution. Once we show Geordi a demonstrably better Bayes solution, he has no justification to not adopt it. **In the restricted context, indecisive agents whose policies are not Bayes solutions can be convinced to become (representable as) Bayesian.**

Visual Intuition



Wald's Complete Class Theorem

Given some plausible assumptions about a statistical decision problem, its set of Bayes solutions is a complete class: every policy outside of it is risk dominated by some policy in it.

Computability Result

Suppose the loss function is computable, the probability distribution over the observable outcomes is computable at each possibility, and the sets of acts, possibilities, and observable outcomes are finite. Then there is a computable procedure that, given a non-Bayes solution, computes the Bayes solution that risk dominates it.

References

[1] Abraham Wald. An essentially complete class of admissible decision functions. *The Annals of Mathematical Statistics*, pages 549–555, 1947.

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